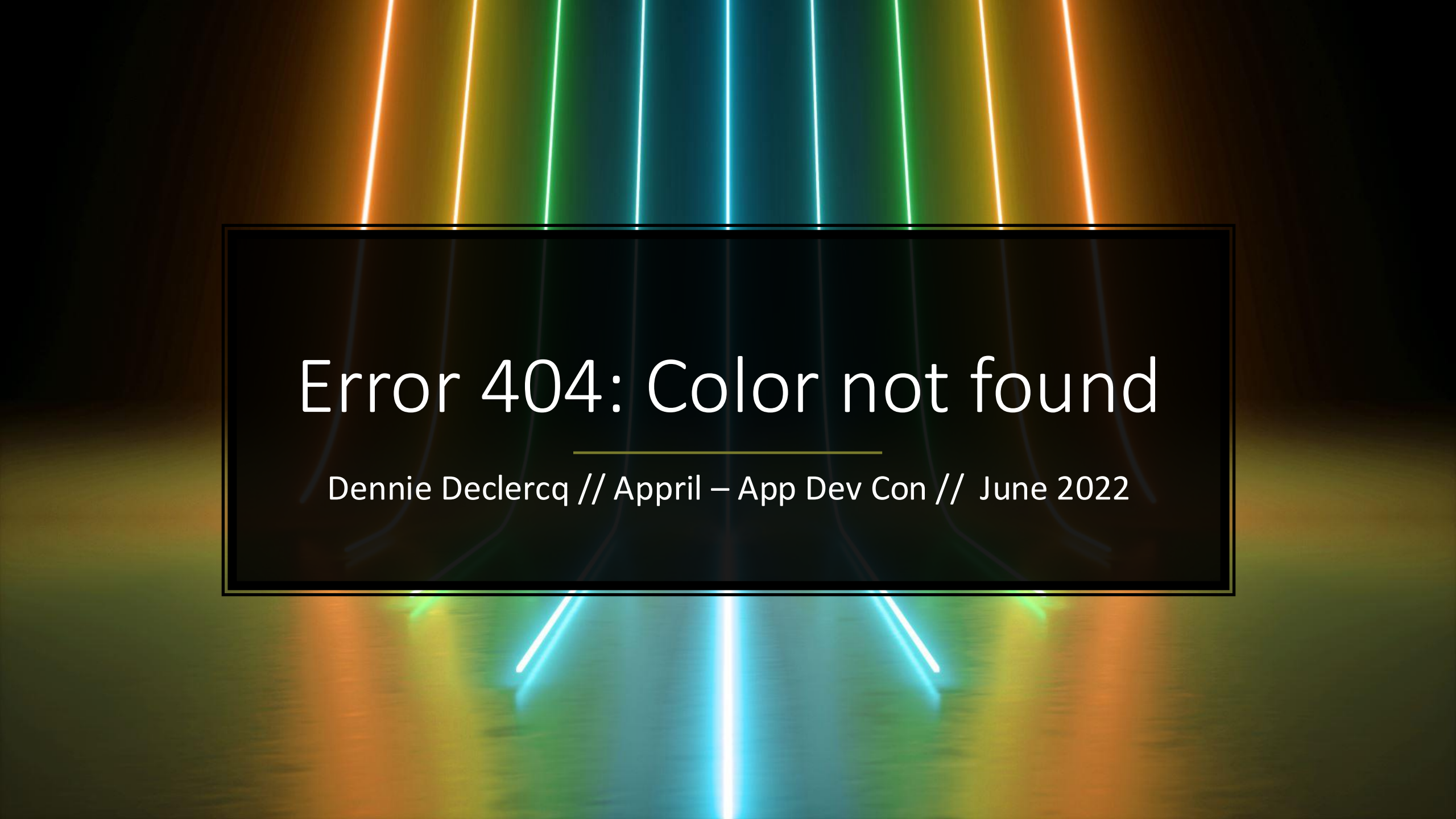


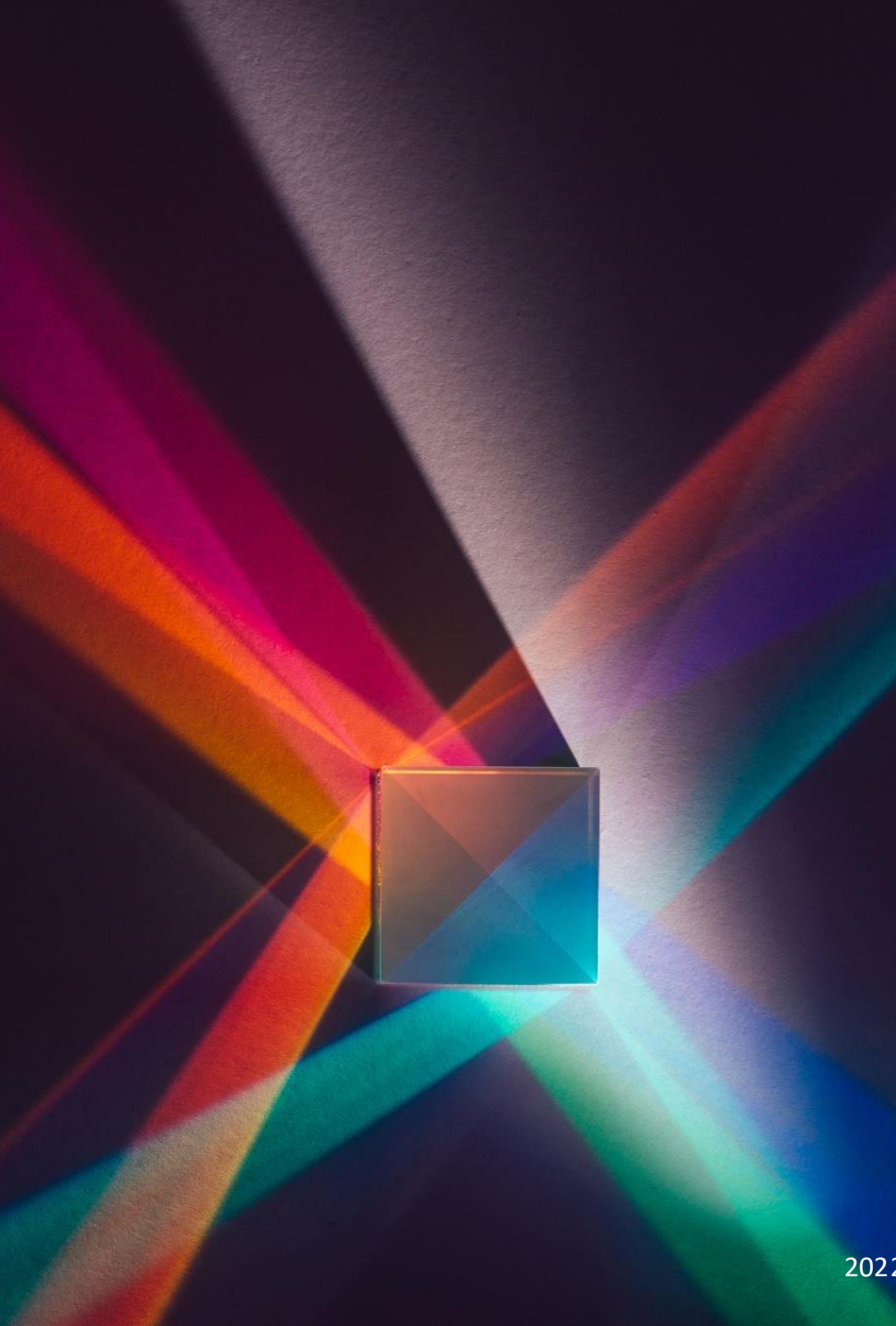


Error 404: Color not found

Dennie Declercq // April – App Dev Con // June 2022



Intro

An abstract background on the left side of the slide. It features a dark, textured surface with several bright, overlapping light rays in shades of orange, red, yellow, green, and blue. These rays appear to emanate from a central point and spread outwards. In the center of this light pattern is a small, semi-transparent square with a blue and white geometric design inside it.

Content

- Intro
- Intro to Accessibility
- Color Contrast
- Combination: Vision & Learning difficulties
- Outro

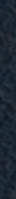


Who am I?

- Dennie Declercq
- President, IT-Coach & Developer DDSoft
- Microsoft MVP
- #DreamingIsBelieving
- @DennieDeclercq



Intro to Accessibility





What is Accessibility?

- People with all needs can access:
 - Internet
 - Daily life activities
 - All areas

Visual Impairment on Wikipedia

Visual impairment, also known as vision impairment, is a medical definition primarily measured based on an individual's better eye visual acuity; in the absence of treatment such as correctable eyewear, assistive devices, and medical treatment—visual impairment may cause the individual difficulties with normal daily tasks including reading and walking. Low vision is a functional definition of visual impairment that is chronic, uncorrectable with treatment or correctable lenses, and impacts daily living. As such low vision can be used as a disability metric and varies based on an individual's experience, environmental demands, accommodations, and access to services. The American Academy of Ophthalmology defines visual impairment as the best-corrected visual acuity of less than 20/40 in the better eye, and the World Health Organization defines it as a presenting acuity of less than 6/12 in the better eye. The term blindness is used for complete or nearly complete vision loss.

Color blindness on Wikipedia

Color blindness (color vision deficiency) is the decreased ability to see color or differences in color. It can impair tasks such as selecting ripe fruit, choosing clothing, and reading traffic lights. Color blindness may make some academic activities more difficult. However, issues are generally minor, and the colorblind automatically develop adaptations and coping mechanisms. People with total color blindness (achromatopsia) may also be uncomfortable in bright environments and have decreased visual acuity.

Color Deficiencies : Protanopia

Normal vision



Protanopia



© Images: <https://www.colourblindawareness.org>

Color Deficiencies: Tritanopia

Normal vision



Tritanopia



© Images: <https://www.colourblindawareness.org>

Color Deficiencies: Deutranopia

Normal vision



Deutranopia



© Images: <https://www.colourblindawareness.org>

Color Deficiencies: Monochromacy

Normal vision



Monochromacy



© Images: <https://www.colourblindawareness.org>

Learning Disabilities on Wikipedia

Learning disability, learning disorder, or learning difficulty (British English) is a condition in the brain that causes difficulties comprehending or processing information and can be caused by several different factors. Given the "difficulty learning in a typical manner", this does not exclude the ability to learn in a different manner. Therefore, some people can be more accurately described as having a "learning difference", thus avoiding any misconception of being disabled with a lack of ability to learn and possible negative stereotyping. In the United Kingdom, the term "learning disability" generally refers to an intellectual disability, while difficulties such as dyslexia and dyspraxia are usually referred to as "learning difficulties".



The importance of Accessibility & Inclusion

- **Inclusion** = All people are living together in the same environment.
- When people depend on others to have access for their needs, they are **excluded**.
- If your communication* is not accessible, then it leads to **exclusion**.
- * Communication examples: websites, apps, leaflets, commercials, banners, posters, (info)graphics.

Covid19 increased exclusion

DIGITAL EXCLUSION

- Website is not accessible.
- App is not accessible.
- Not being able to video conference.
- Not being able to search on YouTube.

PHYSICAL EXCLUSION

- Not being able to enter a building (ex: Because you don't have a specific app).
- Not being able to go outside.
- Not knowing what and when you are allowed to do.

The business side

If your commercial is not accessible for all people, than they can't buy your product.

An inaccessible website can prevent people to buy the product/ service.

1 on 3 men are colorblind.

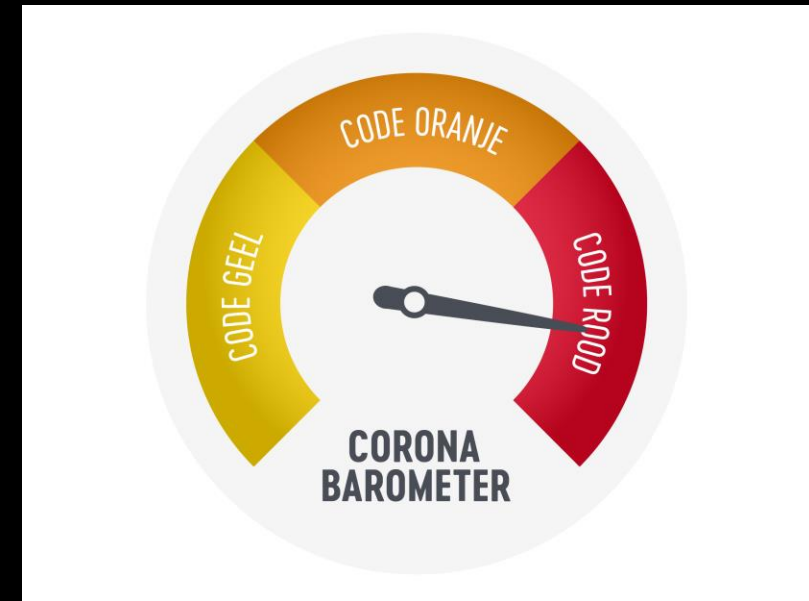
A lot of people (with elderly) have a visual impairment

“Even I look until my eyes hurt on a digital or physical folder!”-Dennie

Hi there ! Insta Girl!

- A lot of modern influencers are making inaccessible content.
- Too many insta style designs, by example on canva.com are not accessible.
- There is an ableism and “far from my bed show” happening.
- With a more accessible design they can reach more people.

(BAD) Examples





Color contrast

What WCAG Says (2.1)

- 1.4.1: Use of Color – Level A
 - Color is not used as the only visual means of conveying information, indicating an action, prompting a response, or distinguishing a visual element.
- 1.4.3: Contrast (Minimum) – Level AA
 - The visual presentation of text and images of text has a contrast ratio of at least **4.5:1**, except for the following:
 - **Large Text:** Large-scale text and images of large-scale text have a contrast ratio of at least **3:1** (min 18 pt or 14 pt bold);
 - **Incidental:** Text or images of text that are part of an inactive user interface component, that are pure decoration, that are not visible to anyone, or that are part of a picture that contains significant other visual content, have no contrast requirement.
 - **Logotypes:** Text that is part of a logo or brand name has no contrast requirements.
- 1.4.6: Contrast (Enhanced) – Level AAA
 - The visual presentation of text and images of text has a contrast ratio of at least **7:1**, except for the following:
 - **Large Text:** Large-scale text and images of large-scale text have a contrast ratio of at least **4.5:1**;
 - **Incidental:** Text or images of text that are part of an inactive user interface component, that are pure decoration, that are not visible to anyone, or that are part of a picture that contains significant other visual content, have no contrast requirement.
 - **Logotypes:** Text that is part of a logo or brand name has no contrast requirement
- 1.4.11: Non-text Contrast – Level AA
 - The visual presentation of the following have a contrast ratio of at least **3:1** against adjacent color(s):
 - **User Interface Components:** Visual information required to identify user interface components and states, except for inactive components or where the appearance of the component is determined by the user agent and not modified by the author;
 - **Graphical Objects:** Parts of graphics required to understand the content, except when a particular presentation of graphics is essential to the information being conveyed.

Contrast Ratio

Important Ratio's

- 3:1
- 4.5:1
- 7:1

Formula

$(L1 + 0.05) / (L2 + 0.05)$, where

L1 is the relative luminance of the lighter of the colors, and

L2 is the relative luminance of the darker of the colors.

Testing for Color Contrast:

Adobe Color Contrast
Analyzer

Microsoft Accessibility
Insights

Tools on MS Windows
(10)

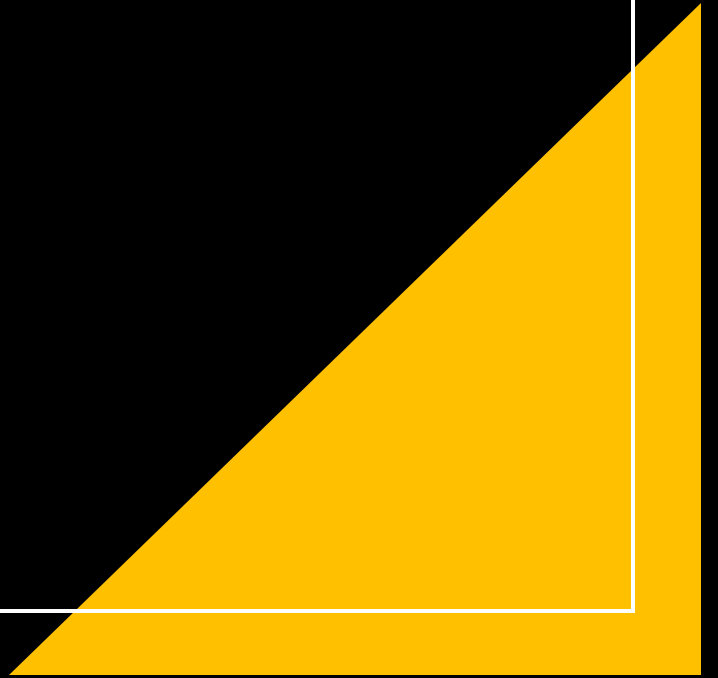
DEMO: Adobe Color Contrast Analyzer

<https://color.adobe.com/nl/create/color-contrast-analyzer>

DEMO: Microsoft Accessibility Insights

<https://accessibilityinsights.io>

DEMO: Ease of access tools on Windows (10): Color Filters





Combination: Vision & Learning difficulties

Basic guidelines

- Color is not the only way of conveying information
- Combine multiple output form factors
- Recognize exclusion
- Provide options
- Look from a child prespective

Red & Green

RED

RED

DANGER

BAD

ALERT

GREEN

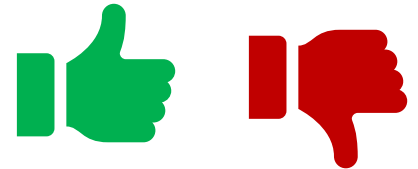
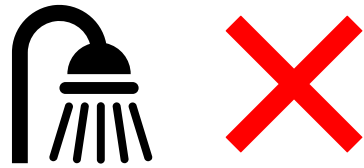
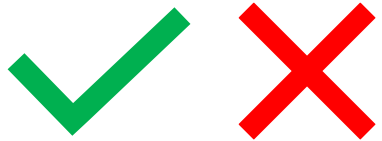
GREEN

SAFE

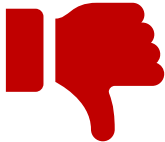
GOOD

CONFIRM

Images

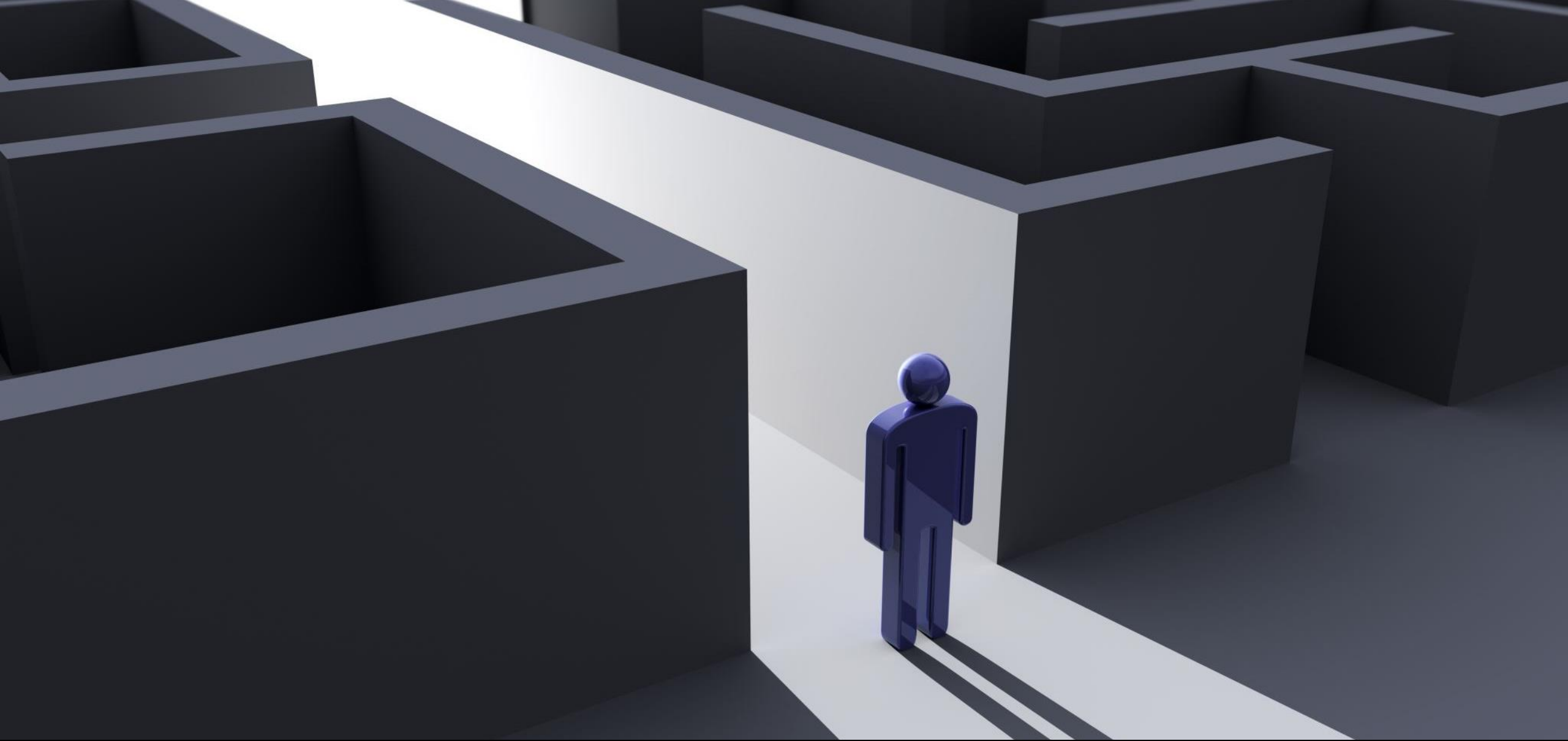


Combining Text & Images

							
Confirm	Cancel	No Shower!		You can shower		Good	Bad

The Holy 3-Unit

- Text
- Color
- Speech



Outro

Recap

- I told about Accessibility & Inclusion
- I gave insights in Color Vision Deficiencies (Color Blindness)
- I showed tools how to handle accessible colors
- I told about designing for people on the color blind spectrum & learning disabilities spectrum.



Thanks || Q&A

- Dennie Declercq
- Dennie@DDSoft.be
- @DennieDeclercq